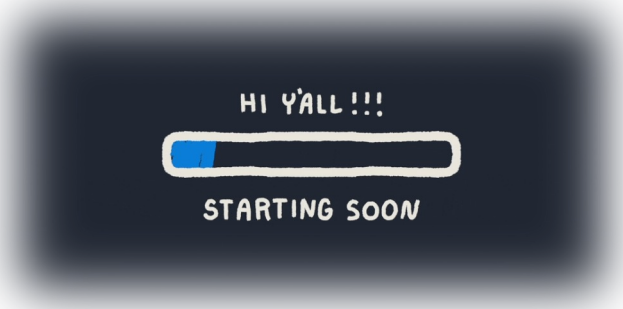
Hypothesis Testing

Trainer: Dr. Darshan Ingle





Another example on Hypothesis Testing

The average IQ of the adult population is 100.

A researcher believes the average IQ of adults is lower.

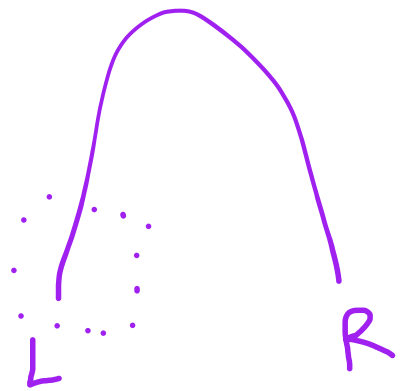
A random sample of 5 adults are tested and scored

→ 69, 79, 89, 99, 109. (s.d. = 15.81)

Is there enough evidence to suggest the average IQ is lower?

(a) $\mu = 100$ H_0

(b) $\mu < 100$ H_1



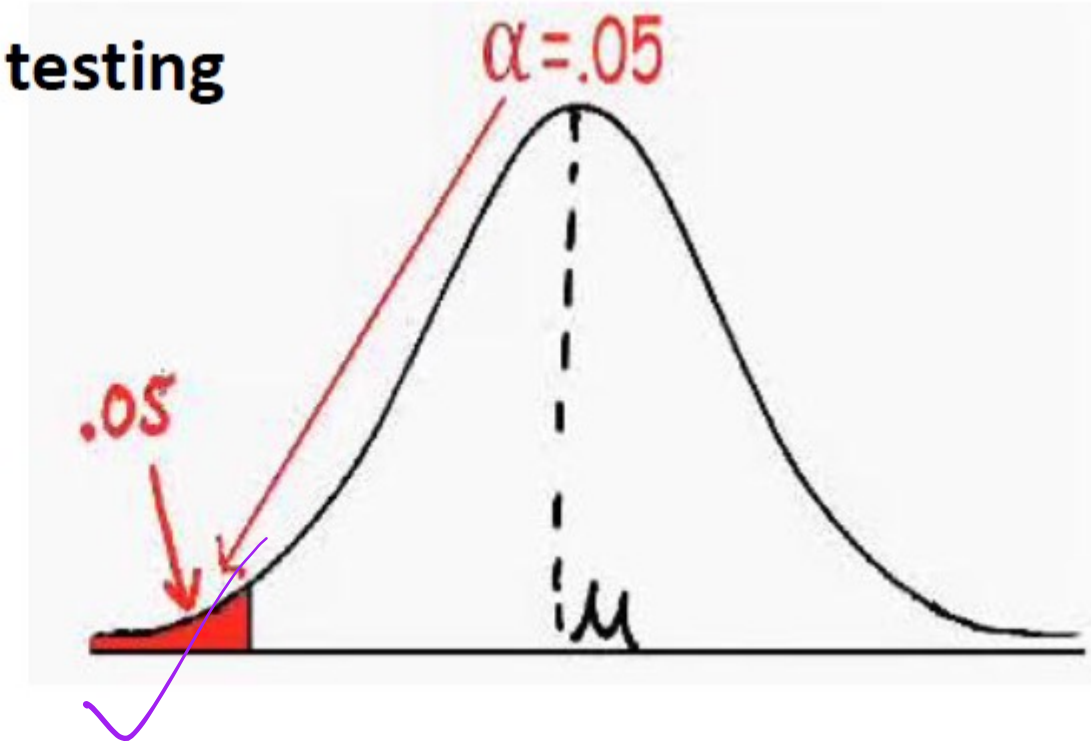
✓ **Recall : 5 steps for Hypothesis testing**

✓ **understand the data given :**

$H_0: \mu = 100$

$H_1: \mu < 100$

One-tailed test



Step 3 : find Critical Values

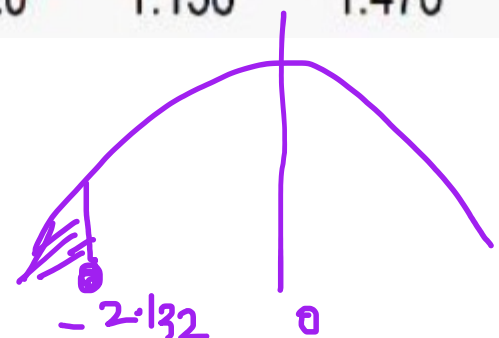
Here we need find z-value or t-value.

When to use t-values

1. σ is unknown
2. Sample size is less than 30

t-table

	area						
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05
df							
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571



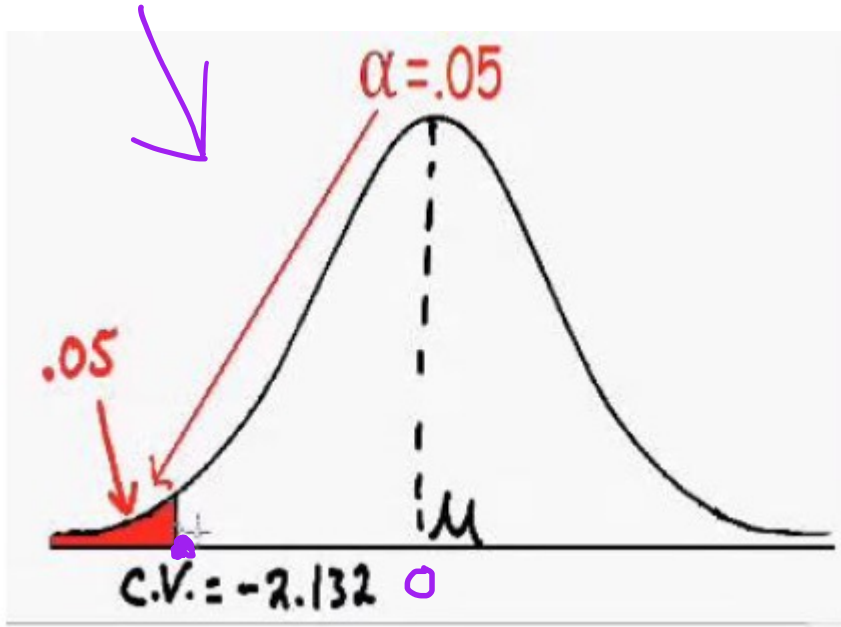
df = degrees of freedom = (sampleSize - 1)

$= n - 1$

$= 5 - 1$

$= 4$

the **C.V** : Critical value on the left is **-ve**
and the **C.V** on the Right is **+ve**



Step 4 : find test statistic. *i.e. the t-value in this case*

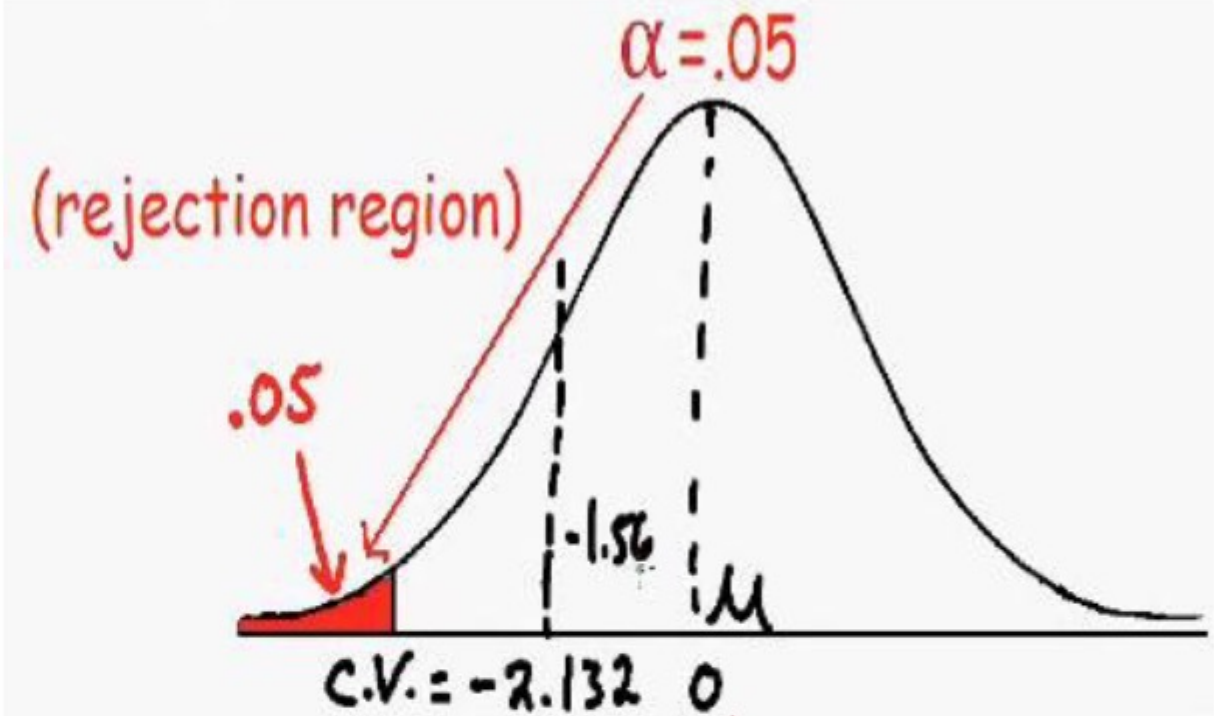
$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{89 - 100}{15.81/\sqrt{5}} = -1.56$$

test statistic

Step 5 : Conclusion

Accept $H_0: \mu = 100$



center point of the Normal Distribution is always t = 0 or z=0

Hence, we say with 95 % confidence that there is enough evidence to support the claim : "Avg IQ =100" or we lack evidence to support "Lower IQ's"

Important Takeaways

How to make hypothesis { Null & Alternative } from a problem statement ?

[understand here what statistics can and cannot do !!]

1. Statistics cannot prove something true or right.
2. It can only prove a assumption or hypothesis as false.

   ex: in court you are either Not guilty or guilty ; its never innocent.
3. Claims or Hypothesis can be of 2 types :

   a> based on proportion

   b> based on mean

   Ex. Most people get Jobs thru Networking.

   Ex. The average payload of trucks on NH is 1.2 tons.

Important Takeaways

4. H0 , null hypo should always have a =≤ ≥

   H1 , alternative hypo would have <, >, ≥, ≤ or !=

5. Understand How to make define Hypothesis or a claim thru below examples.

   Remember : to start by assuming that your H0 as true

Ex1: The mean of the fluid is at least 100ml in a can.

Write the claim

$\mu \geq 100$ $\mu < 100$

Write the opp.

claim :	μ	>=100	As = here , hence it is H_0
Opp. :	μ	<100	

Important Takeaways

Ex1: Most CEOs are Male

claim : p > 0.5 (proportion)

Opp. : p <= 0.5

As = here , hence it is H0

H_0 : p = 0.5

H_1 : p > 0.5

Thereby, we would the Hypothesis Testing to Reject or not to reject H_0

Important Takeaways

Note:

- **To figure out H0 and H1 we need to have understand concept of claim and opp.**

Claim can be on proportion

like "most of the people" or "at-least"

or on mean

like average weight of Indian hockey players is 59.8 kgs

z-value vs. t-value calculation or z-statistic vs. t-statistic

When to use t-values

1. σ is unknown
2. Sample size is less than 30

&

Ex1: The average test score for an entire school is 75 with a standard deviation of 10. What is the probability that a random sample of 5 students scored above 80?

μ σ n $\bar{x}$

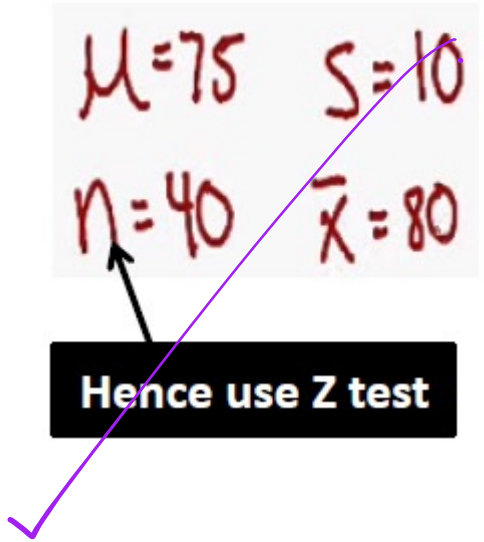
$\mu = 75$ $\sigma = 10$ $n = 5$ $\bar{x} = 80$

Hence use Z test

z-value vs. t-value calculation or z-statistic vs. t-statistic

Ex2:

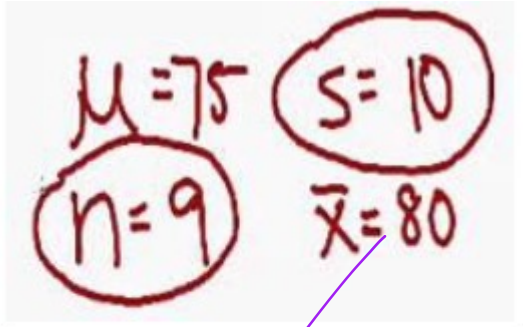
The average test score for an entire school is 75. The standard deviation of a random sample of 40 students is 10. What is the probabilty the average test score for the sample is above 80?



z-value vs. t-value calculation or z-statistic vs. t-statistic

Ex3:

The average test score for an entire school is 75. The standard deviation of a random sample of 9 students is 10. What is the probabilty the average test score for the sample is above 80?



Both conditions met. Hence use t-test

$p > 0.50$ $p \leq 0.50$ $\frac{118}{200} = 0.59$

Hypothesis Testing – portion i.e. no mean value

The NFL believes the chance of the Patriots winning the coin toss is greater than 50%. In a random sample of 200 coin tosses, the Patriots won 118 times. Is there enough evidence to suggest the Patriots are cheaters?

1. State null (H_0) and alternative (H_1) hypothesis
2. Choose level of significance (α)
3. Find critical values
4. Find test statistic
5. Draw your conclusion

Step 1:

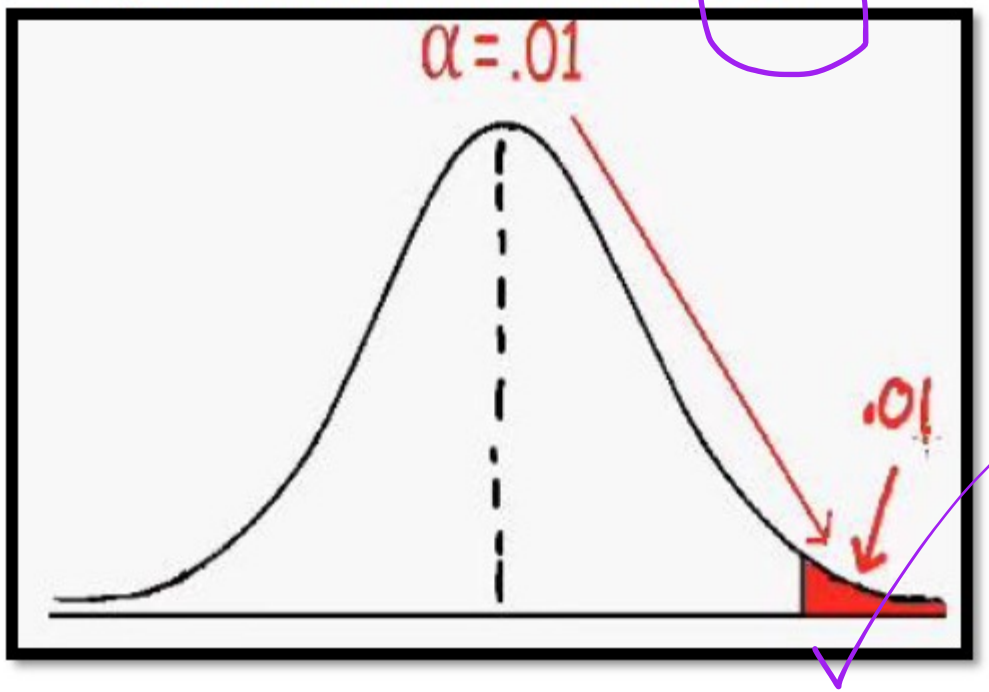
$H_0: p \leq .50$

(1 tailed test)

$H_1: p > .50$

Step 2: Choose Confidence level of 95 or 99%

Or significance level of 5% or 1%.

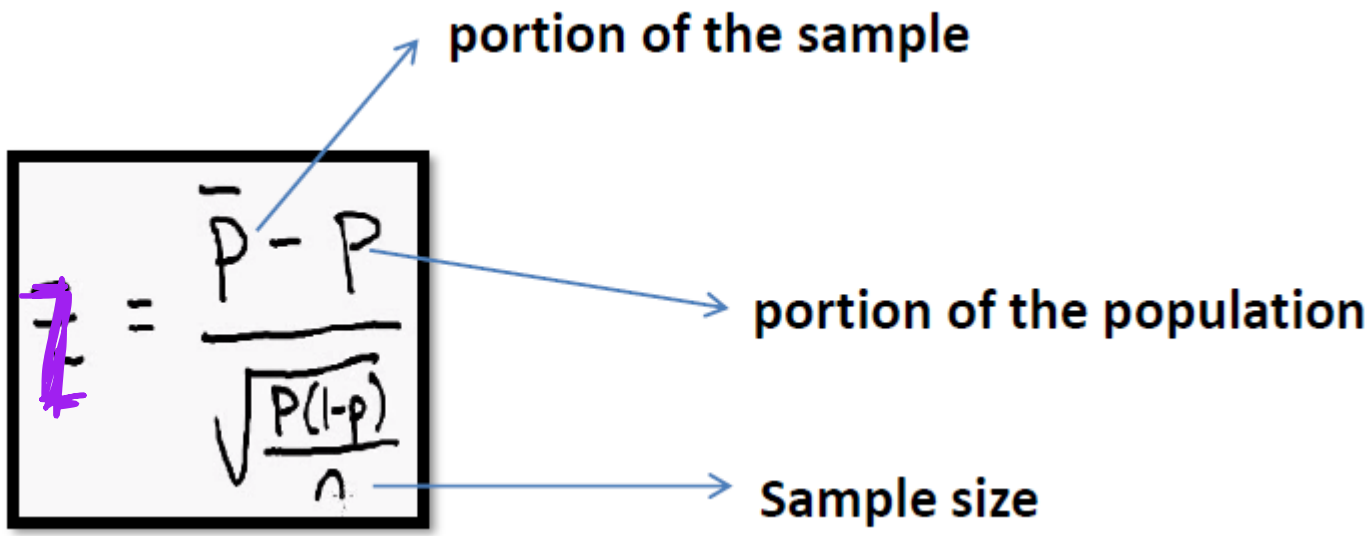


Step 3: C.V w.r.t portion is always the z-value.

Z-value from z-table = 2.326

Step 4: find test statistic

I.e. z-value.



Note : different formulae

$\bar{p}$ = 118 / 200 = .59

$$\frac{.59 - .50}{\sqrt{\frac{(.5)(.5)}{200}}} = 2.55$$

Step 5

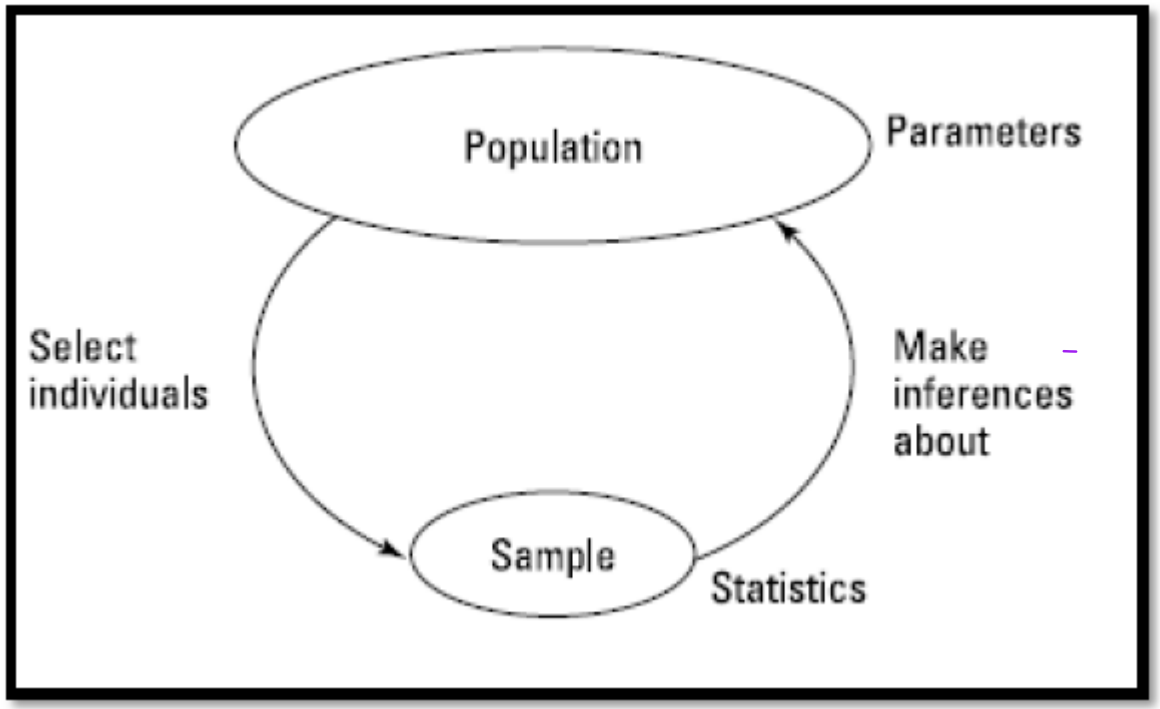
As the test statistic = 2.55 is > 2.36 , hence it falls in the Rejection Ratio.

This means we would reject the H_0

i.e. we accept the H_1 i.e. we accept the claim that Patriots are cheaters

Or the coin Patriots use is Not fair

Confidence Intervals



Its not possible most of the times to calculate the population parameters. **That's where Statistics help**. We take a sample , calculate all sample parameters and then make estimates about the population parameters with **some confidence**. { **may be 99% or 95%** }

Confidence Intervals

We want to estimate the average weight for the adult male population. The average weight of 100 randomly selected adult males is 180 lbs. Assume a population standard deviation of 20 lbs. Compute a 95% confidence interval for the population average weight.

Ex:

Population mean = Sample mean (not ok)

$\bar{x} \pm e$

Confidence Interval → Population mean = Sample mean +/- sample error

Confidence Intervals – with z-value

We want to estimate the average weight for the adult male population. The average weight of 100 randomly selected adult males is 180 lbs. Assume a population standard deviation of 20 lbs. Compute a 95% confidence interval for the population average weight.

Step 1 : **Analyze the data and decide whether we would use t-value or z-value.**

Ans: would use z-value



When to use t-values

1. σ is unknown
2. Sample size is less than 30

Confidence Intervals – with z-value

Step 2 : Compute. → Population mean = Sample mean +/- sample error

$$\mu = \bar{X} \pm \frac{z \cdot \sigma}{\sqrt{n}}$$

Refer z-table for 95% confidence interval or 5% significance level...

Confidence Intervals – with z-value

Z-table

Confidence Level	Area between 0 and z-score	Area in one tail (alpha/2)	z-score
50%	0.2500	0.2500	0.674
80%	0.4000	0.1000	1.282
90%	0.4500	0.0500	1.645
95%	0.4750	0.0250	1.960
98%	0.4900	0.0100	2.326
99%	0.4950	0.0050	2.576

Confidence Intervals – with z-value

$$\mu = 180 \pm \frac{1.96 \cdot 20}{\sqrt{100}}$$

$$\mu = 180 \pm 3.92$$

$$\mu = (176.08, 183.92)$$ 95%

Confidence Intervals – with t-value

The test scores of 9 randomly selected students are

83, 73, 62, 63, 71, 77, 77, 59, 92.

Compute the 99% confidence interval of the true mean.

Step 1 : Analyze the data and decide whether we would use t-value or z-value.

Ans: would use t-value



When to use t-values

1. σ is unknown
2. Sample size is less than 30

Confidence Intervals – with t-value

Step 2 : Compute. → Population mean = Sample mean +/- sample error

$$\mu = \bar{x} \pm \frac{t \cdot s}{\sqrt{n}}$$

Refer t-table for 99% confidence interval or 1% significance level…

Confidence Intervals – with t-value

$$\mu = 73 \pm \frac{(3.355)(10.69)}{\sqrt{9}}$$

$$\mu = 73 \pm 11.95$$

$$\mu = (73 - 11.95 \;,\; 73 + 11.95)$$

$$\mu = (61.05 \;,\; 84.95)$$

	Confidence Interval						
	80%	90%	95%	98%	99%	99.8%	99.9%
	α level two-tailed test						
df	0.2	0.1	0.05	0.02	0.01	0.002	0.001
1	3.078	6.314	12.706	31.821	63.657	318.313	636.589
2	1.886	2.920	4.303	6.965	9.925	22.327	31.598
3	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	1.372	1.812	2.228	2.764	3.169	4.144	4.587

degree of freedom = (sampleSize - 1)

Final Ans : Hence , we can say with 99% confidence that our population mean is in between { 61.05 and 84.95 }

Chi-Square Test (also called Goodness of Fit Test)

Purpose :

Chi Square statistics are formulated to determine whether an observed number differs either from chance or from what was expected.

Pearson had termed it as a **Goodness-of-fit** test to describe how well theoretically the observed frequencies fit the expected frequencies.

Groups and Numbers

You research two groups and put them in categories single, married or divorced:

The numbers are definitely different, but ...

- Is that just random chance?
- Or have you found something interesting?

The **Chi-Square Test** gives a "p" value to help you decide!

Note : It is one form of Hypothesis test for a Group of data.

Understand Chi-Square Test through an Example

A Casino , states that following is the way die throws show up =>

Could you figure out, whether the Casino is using a Fair Die or Not ?

204

Observed						
#	1	2	3	4	5	6
freq.	22	24	38	30	46	44

// we would follow same 5 steps of Hypothesis testing. before that , fill in your expected values.

total throws = 204

E(1) = (1/6) * 204 = 34

E(2) = (1/6) * 204 = 34

....

.....

Expected						
#	1	2	3	4	5	6
freq.	34	34	34	34	34	34

Step 1> Define H_0

H_0 : Die is fair

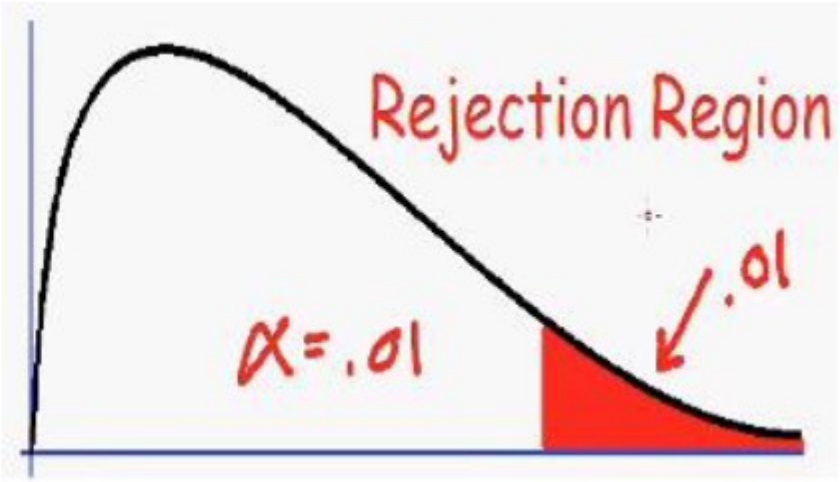
H_1 : Die is unfair

means : Observed data fits the expected data for good.

Understand Chi-Square Test through an Example

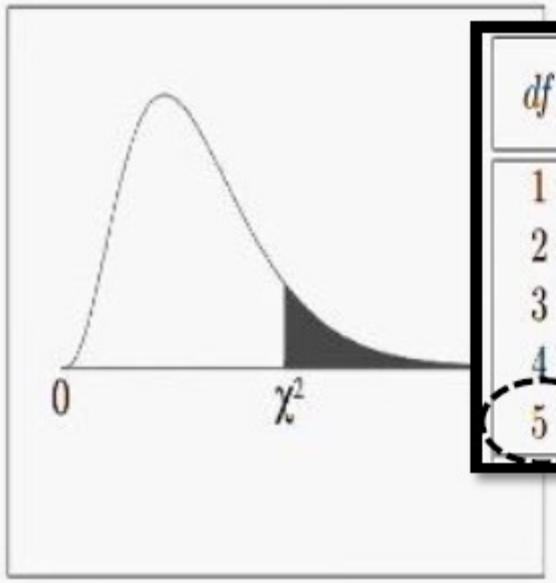
Step 2: Choose the significance level.

5% or 1% => 0.05 or 0.01



Understand Chi-Square Test through an Example

Step 3: Compute the Critical value. Use the Chi-Squared Table. {Not Z / T table}



The shaded area is equal to α for $\chi^2 = \chi^2_\alpha$.

df	$\chi^2_{.995}$	$\chi^2_{.990}$	$\chi^2_{.975}$	$\chi^2_{.950}$	$\chi^2_{.900}$	$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750

df = degree of freedom = (possible outcomes – 1)

df = 6 – 1 = 5

Understand Chi-Square Test through an Example

Step 4: Compute the Chi-squared test statistic

$$X^2 = \text{Sum}\left[\frac{(O-E)^2}{E}\right]$$

$$X^2 = \frac{(22-34)^2}{34} + \frac{(14-34)^2}{34} \ldots\ldots$$

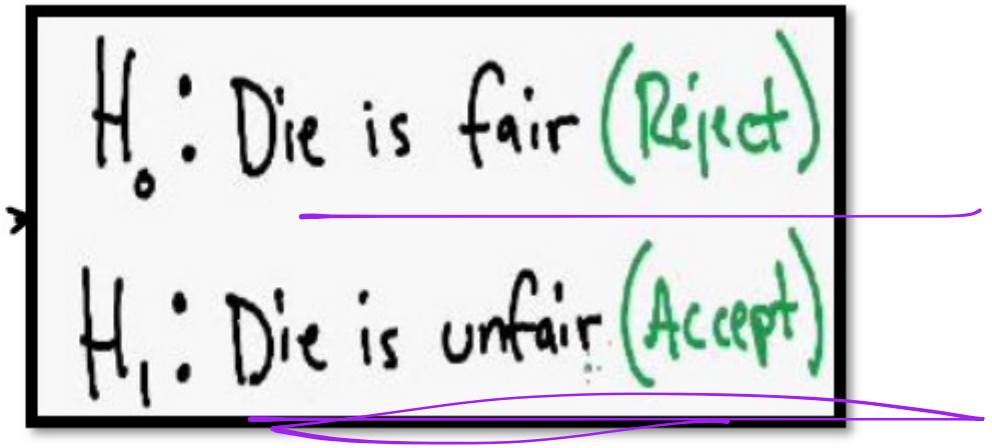
we get the ans. as -----> 15.29 test statistic

Understand Chi-Square Test through an Example

Step 5: Conclusion

As 15.29 > 15.086 , hence test statistic falls in the rejection region.

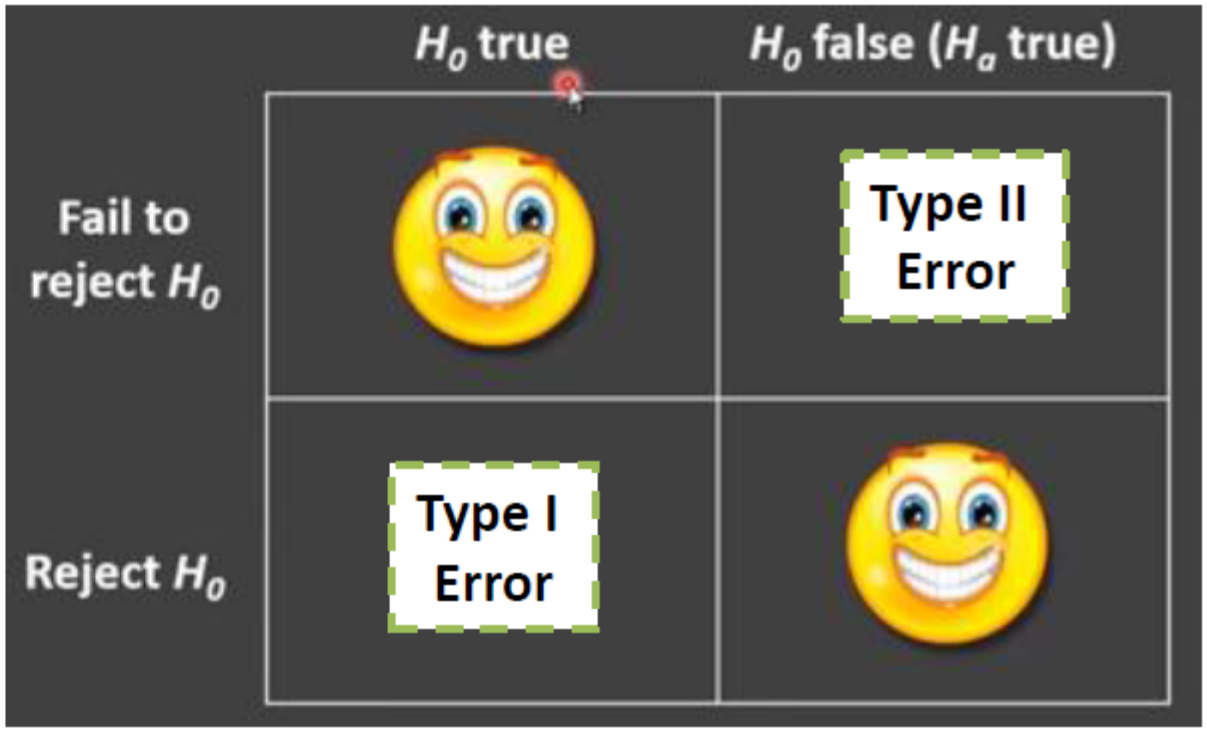
I.E. we can REJECT the H_0



Type I and Type II Error => Similar to False Positive & false Negative*

Truth about the population

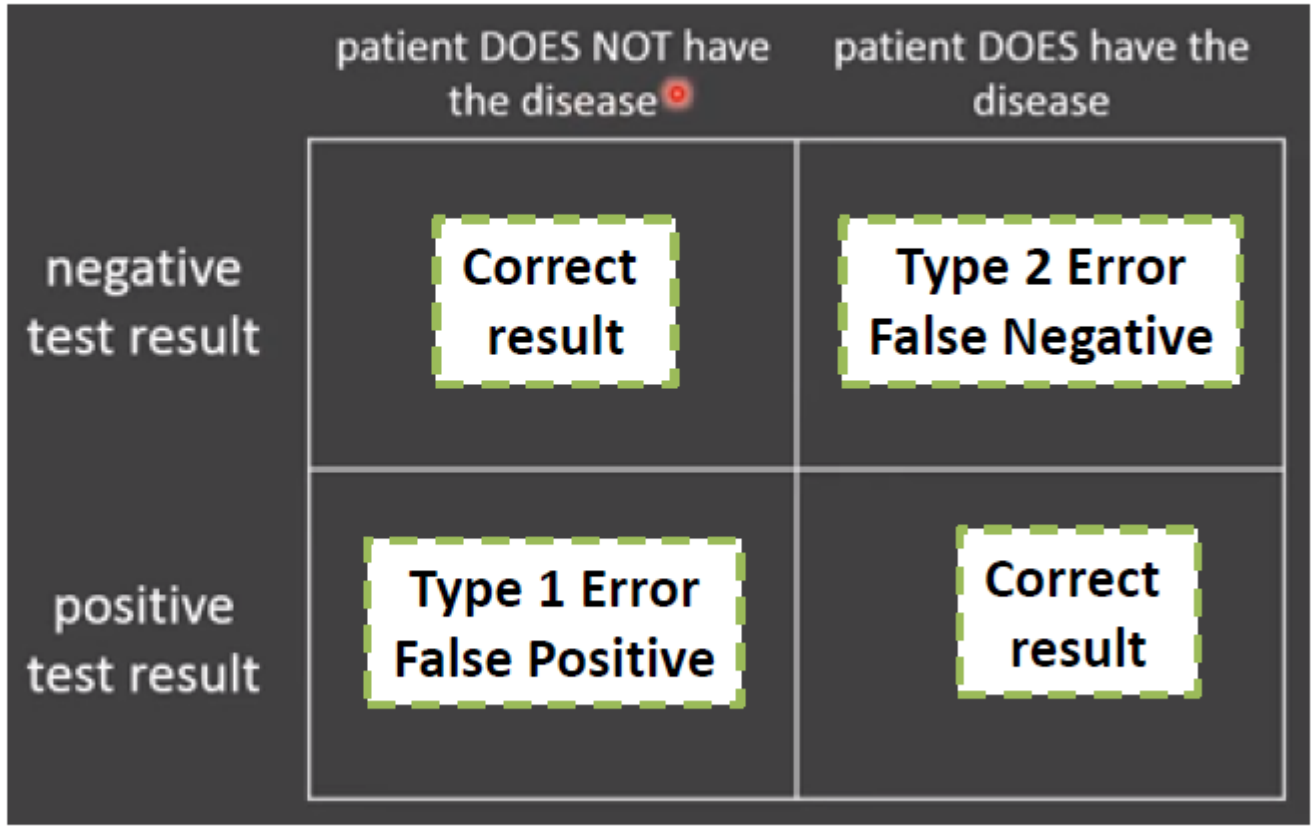
Decision based on the sample



- ✓ Type 1 Error : Reject H_0 when in fact H_0 is true.
- ✓ Type 2 Error : fail to Reject H_0 when in fact H0 is false.{or H_1 is true}

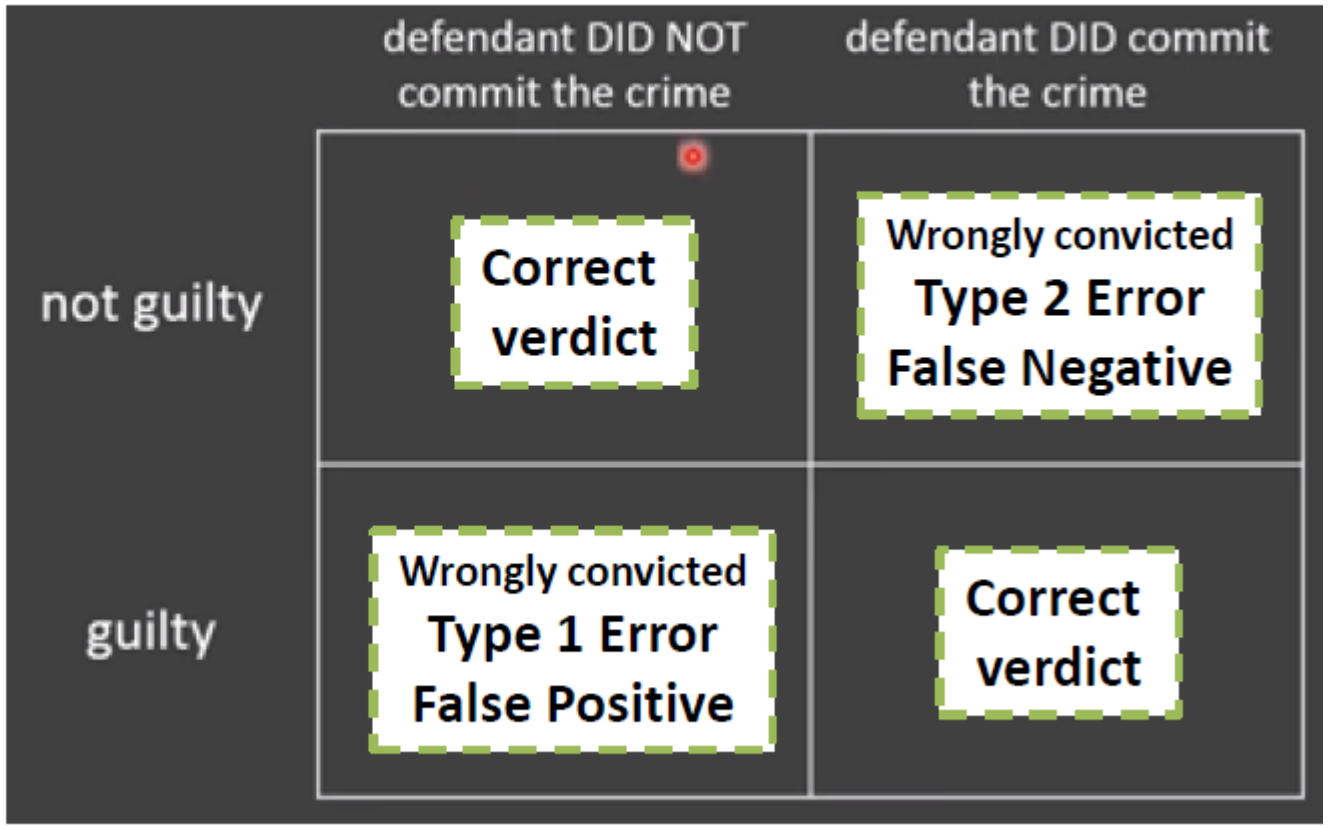
Truth about the patient

Test Results



Truth about the defendant

Verdict



Solve this : Type 1 Error Problem

A large nationwide poll recently showed an unemployment rate of 9% in the US. The mayor of a local town wonders if this national result holds true for her town, so she plans on taking a sample of her residents to see if the unemployment rate is significantly different than 9% in her town.

Let p represent the unemployment rate in her town. Here are the hypotheses she'll use:

$H_0 : p = 0.09$
$H_a : p \neq 0.09$

Type I error - reject H_0 even though H_0 true

Under which of the following conditions would the mayor commit a Type I error?

(A) She concludes the town's unemployment rate is not 9% when it actually is.

Ans : A

(B) She concludes the town's unemployment rate is not 9% when it actually is not.

(C) She concludes the town's unemployment rate is 9% when it actually is.

(D) She concludes the town's unemployment rate is 9% when it actually is not.

Solve this : Type 2 Error Problem

A large university is curious if they should build another cafeteria. They plan to survey a sample of their students to see if there is strong evidence that the proportion interested in a meal plan is higher than 40%, in which case they will consider building a new cafeteria.
Let p represent the proportion of students interested in a meal plan. Here are the hypotheses they'll use:

$H_0 : p \leq 0.40$
$H_a : p > 0.40$

failing to reject H_0, even though it is false

What would be the consequence of a Type II error in this context?

(A) They don't consider building a new cafeteria when they should.

(B) They don't consider building a new cafeteria when they shouldn't.

(C) They consider building a new cafeteria when they shouldn't.

(D) They consider building a new cafeteria when they should.

Ans : A

ANOVA test – Analysis of Variance test

- How do we compare means of many population groups ?

Concept : This is a kind of Hypothesis Testing.

We can compare the **population means** of **2 or more groups.**

Assume

$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$ "k" population

H_a : At least one mean differs from the others.

Objective of ANOVA is to

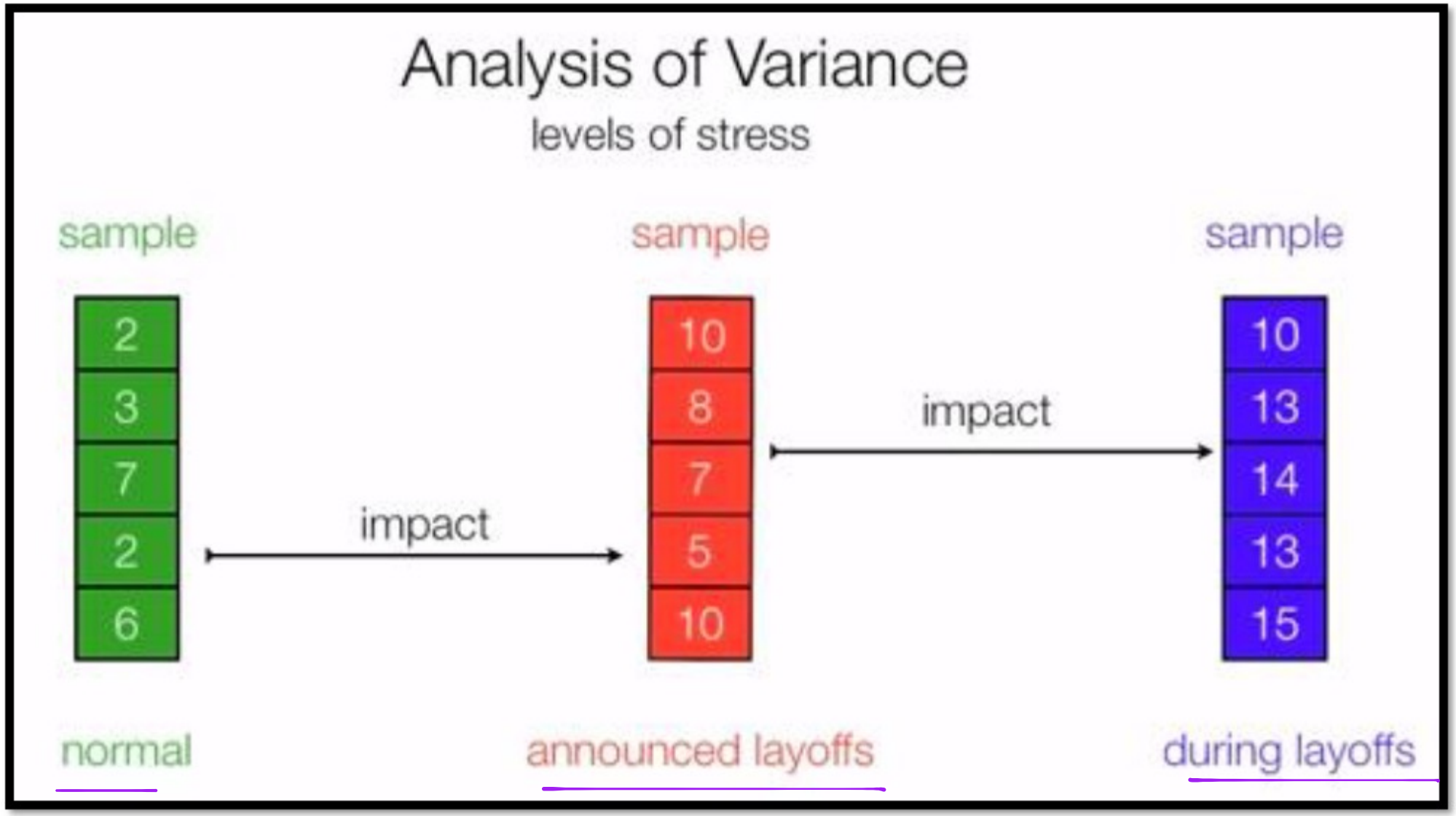
- ❑ REJECT H_0
- ❑ Fail to REJECT H_0

There are other techniques like

- ✓ Independent Samples – t-test
- ✓ Dependent Samples – t-test

Note : Both can compare just two population parameters. Hence ANOVA is more common.

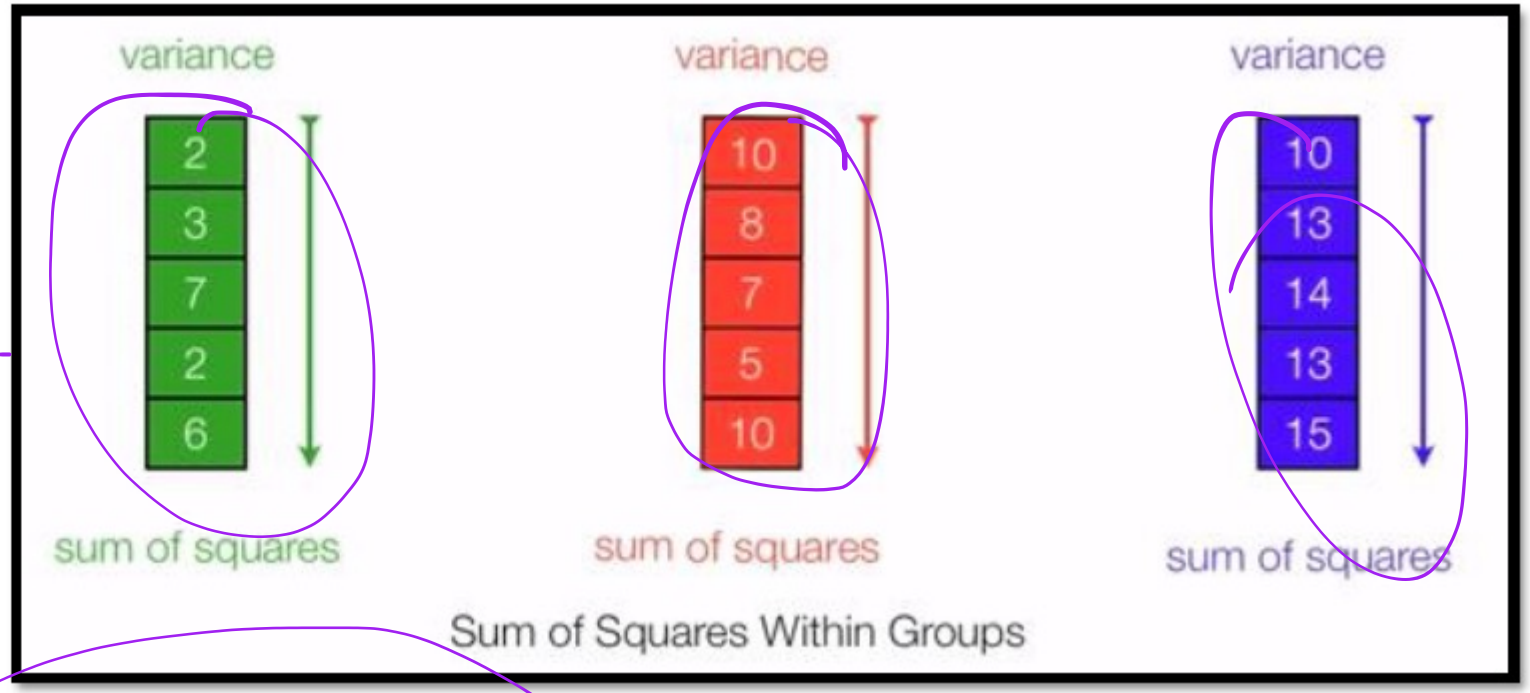
Ex.:

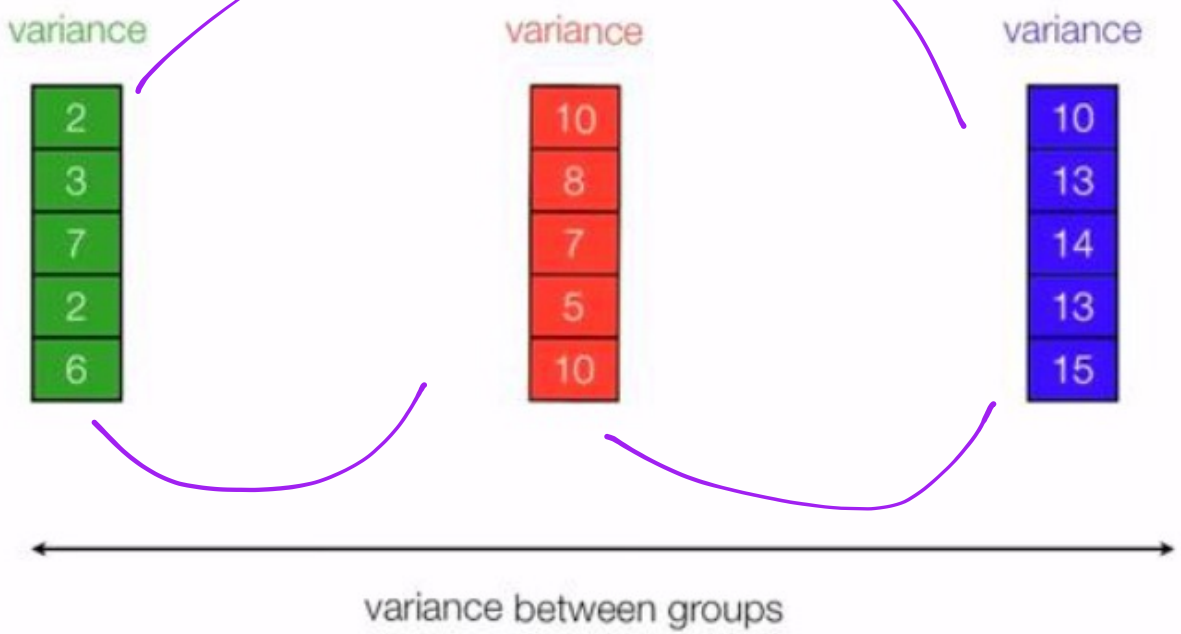


- ✓ **We want to measure "Level of stress" during layoffs.**
- ✓ **We have some observations as above.**

Steps :

1> find the variance within groups
i.e. **s**um of **s**quares **w**ithin Groups
(**called SSW**)

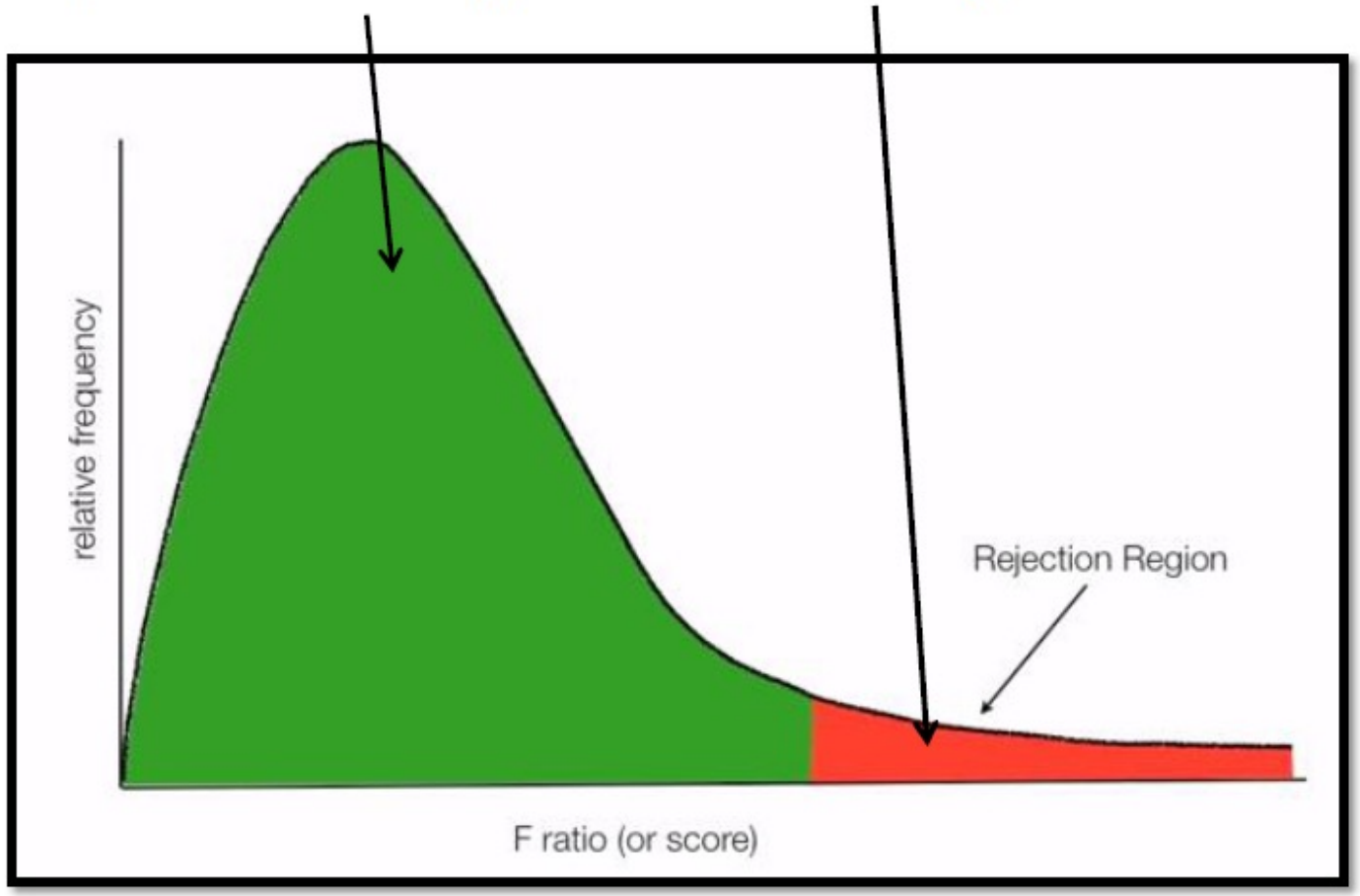




2> find the variance between Groups
i.e. **s**um of **s**quares **b**etween Groups
(**called SSB**)

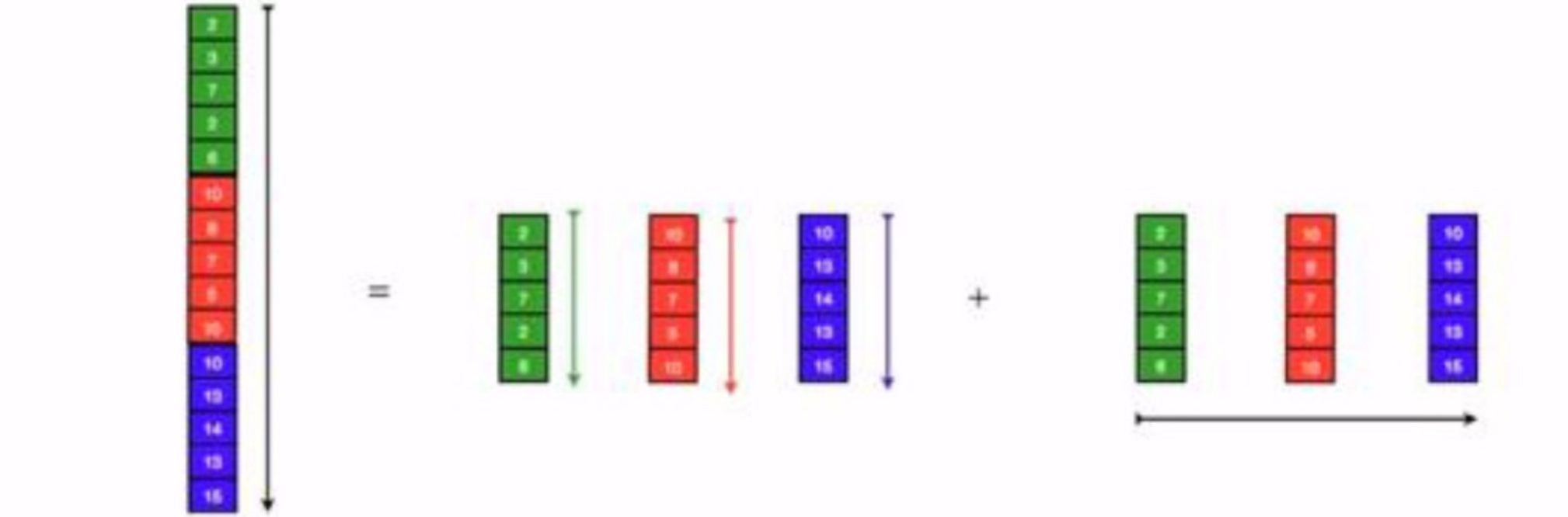
3> Combine all groups together as one single unit and calculate the "total sum of squares" (**called as SST**).

4> Now find the Critical Value i.e. F-Ratio. This is to check whether the F-Ratio lies in the rejection region or not. If it lies in the Rejection region then **REJECT H_0 ,** else Fail to **REJECT H_0**





5> Make following calculation for F-ratio:



6> Compare the F-Ratio w.r.t C.V. to either REJECT or FAIL to REJECT the H_0

Performing Calculation according to Steps :

1> find the variance within groups i.e. sum of squares within Groups (**called SSW**)

sample

2	- 4 =	-2^2	4
3	- 4 =	-1^2	1
7	- 4 =	3^2	9
2	- 4 =	-2^2	4
6	- 4 =	2^2	4
			22

$\sum$ (observation - mean)^2

sample

10	- 8 =	2^2	4
8	- 8 =	0^2	0
7	- 8 =	-1^2	1
5	- 8 =	-3^2	9
10	- 8 =	2^2	4
			18

sample

10	- 13 =	-3^2	9
13	- 13 =	0^2	0
14	- 13 =	1^2	1
13	- 13 =	0^2	0
15	- 13 =	2^2	4
			14

sum of squares within Groups
(called SSW)
= 22 + 18 + 14 = 54

2 & 3> find the variance within groups i.e. sum of squares within Groups (**called SSW**) , by **finding the total sum of squares. Like this :**

observation		mean	observation - mean	(observation - mean) ^{2}
2	-	8.3	= -6.3	40.1
3	-	8.3	= -5.3	28.4
7	-	8.3	= -1.3	1.8
2	-	8.3	= -6.3	40.1
6	-	8.3	= -2.3	5.4
10	-	8.3	= 1.7	2.7
8	-	8.3	= -0.3	0.1
7	-	8.3	= -1.3	1.8
5	-	8.3	= -3.3	11.1
10	-	8.3	= 1.7	2.8
10	-	8.3	= 1.7	2.8
13	-	8.3	= 4.7	21.8
14	-	8.3	= 5.7	32.1
13	-	8.3	= 4.7	21.8
15	-	8.3	= 6.7	44.4
				257.3

5 = 8.3

Total sum of squares (called SST) = 257.3

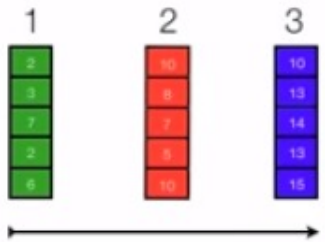
sum of squares between Groups (called SSB) = 257.3 – 54 = 203.3

4 & 5> Finding F-ratio and then comparing with F-table.

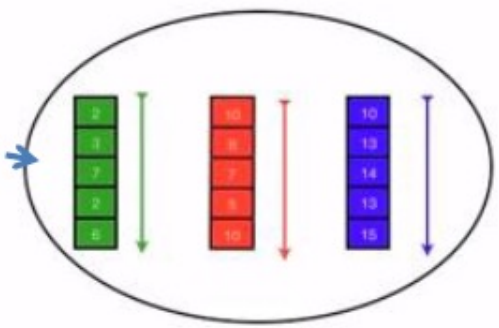
$$\frac{\text{Sum of Squares Between Groups}}{\text{degrees of freedom}} = \frac{203.3}{2} = 101.667$$

Groups - 1

3 - 1 = **2**



$$\frac{\text{Sum of Squares Within Groups}}{\text{degrees of freedom}} = \frac{54}{12} = 4.5$$



15 - 3 = 12

Observations - Groups

Hence we calculate the F-ratio as :

$$F = \frac{101.667}{4.5} = 22.59$$

f(2,12) = 22.59 , p<.05

degree of freedom , Numerator

degree of freedom , Denominator

Final step : look into the F-table and check the calculated F-value > the C.V or not.

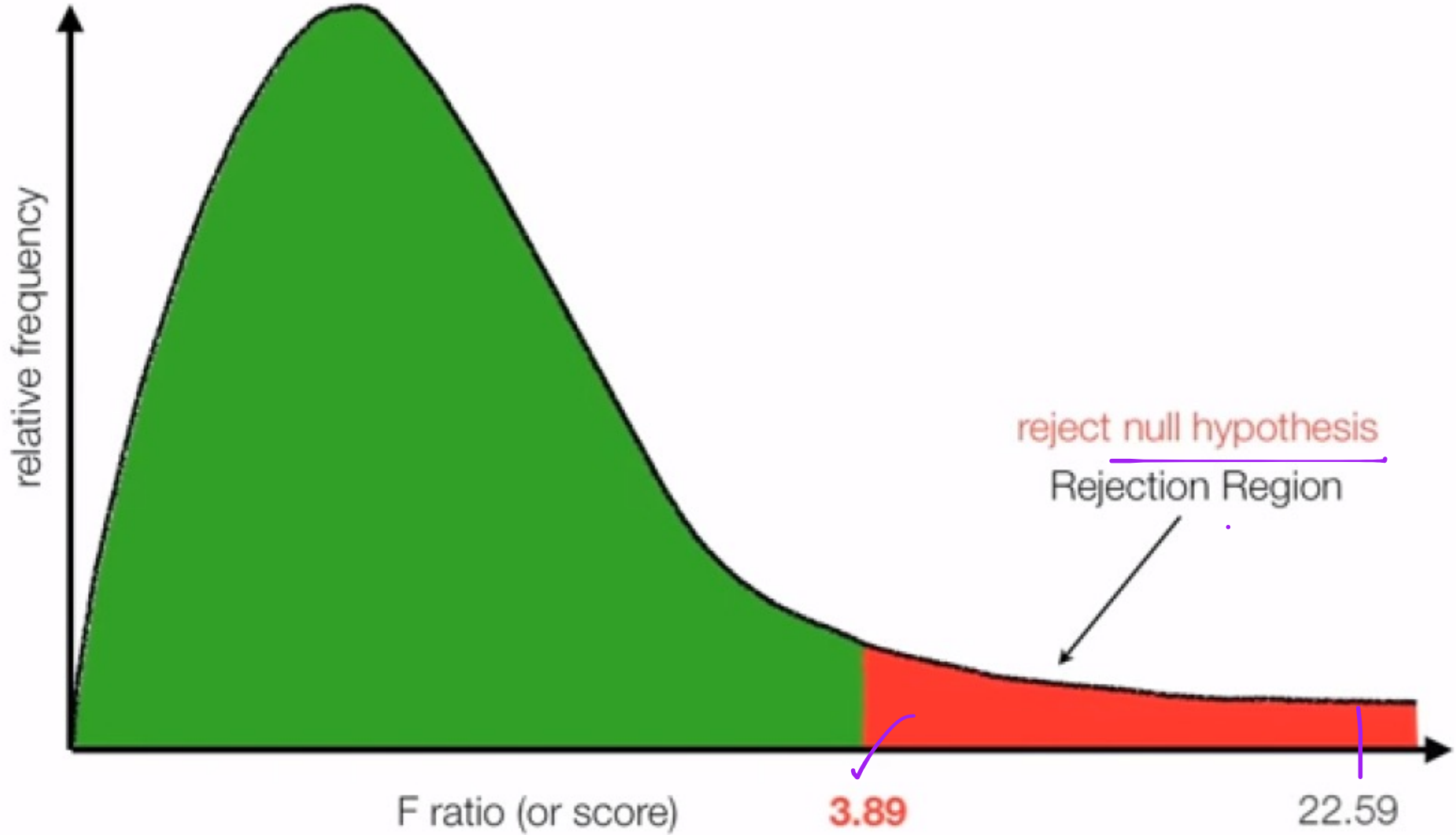
Accordingly conclude to Reject or Not Reject H_0

C.V for f(2,12) = 3.89 // see below F-table

degrees of freedom numerator

degrees of freedom denominator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	161.5	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	246.0	248.0	249.1	250.1
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45	19.45	19.46
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.70	8.66	8.64	8.62
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25

Conclusion : f(2,12) = 22.59 is > than C.V = 3.89 , i.e. the calculated F-value comes in the rejection Region. Hence REJECT H_0 { *see plot on next slide* }



ANOVA test is also called F-Statistic test

Thank You